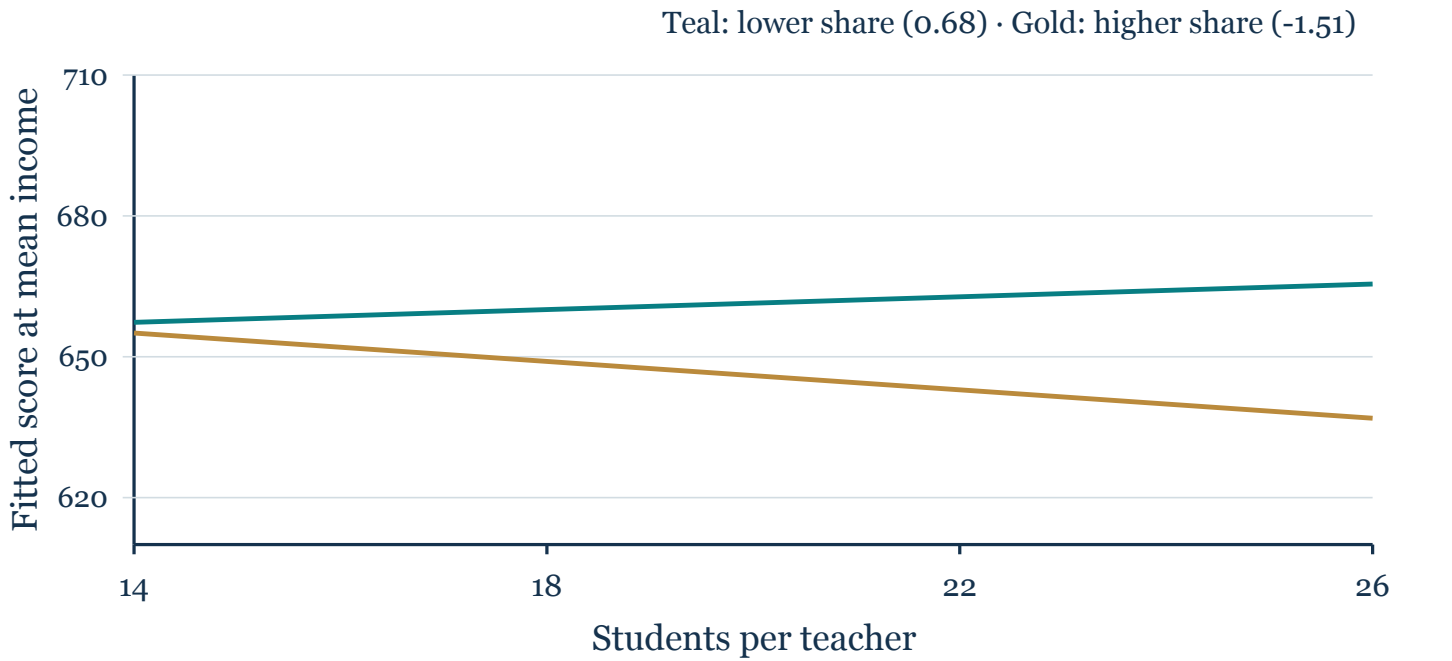
Does the staffing association differ across district groups?

An interaction lets slopes differ in a stated way.



Actual group-specific fitted lines at the overall sample mean of income. High language share means at least 10 percent English learners.

Define the indicator and reference group

Coding determines which coefficients describe the baseline.

> $D_i = 1\{\text{EL}_i \geq 10\}$

Write the interacted equation

Include the main terms as well as their product.

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 D_i + \beta_3 X_i D_i + \gamma W_i + u_i$$

X: staffing; D: group; W: income; XD: their interaction.

Substitute zero and one before interpreting

Two group equations reveal the relevant combinations.

> $D = 0: \quad \partial E(Y \mid X, D, W)/\partial X = \beta_1$

> $D = 1: \quad \partial E(Y \mid X, D, W)/\partial X = \beta_1 + \beta_3$

Interaction contrast

Try the comparison before opening the explanation.

Practice

In $Y = a + 0.4X - 0.9(X \times near) + c\,near + u$, near is 0 or 1. What are the x slopes far from and near the border?

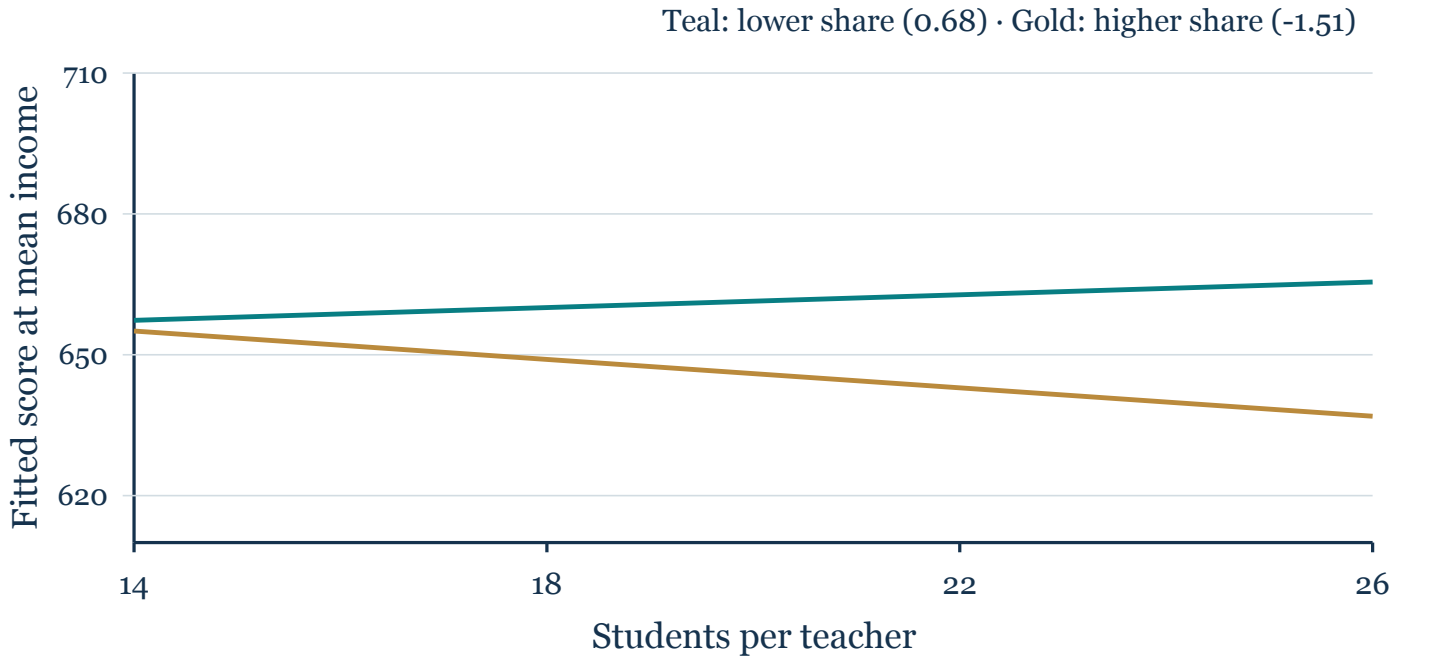
Interaction contrast: explanation

Follow the calculation and interpret its units.

> Far: 0.4. Near: $0.4 - 0.9 = -0.5$. The interaction is the difference in slopes, −0.9. The near-group intercept shift c is a different quantity.

Read both fitted lines in the actual school data

The reference slope and slope difference can have different signs.



Actual group-specific fitted lines at the overall sample mean of income. High language share means at least 10 percent English learners.

Factor-variable syntax keeps the model explicit

Use Stata's interaction notation to include the needed terms.

```
. regress score c.str##i.high income, vce(robust)

------------------------------------------------------------------------------
             |               Robust
       score | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         str |   .6800597   .3532615     1.93   0.055    -.0143453    1.374465
      1.high |   28.41596   12.83889     2.21   0.027     3.178593    53.65332
             |
  high#c.str |
         1   |  -2.192258   .6399264    -3.43   0.001    -3.450159   -.9343566
             |
      income |   1.651672   .1034089    15.97   0.000     1.448401    1.854942
       _cons |   622.5112   7.209389    86.35   0.000     608.3397    636.6826
------------------------------------------------------------------------------
```

Calculate a policy contrast within a group

A two-student change uses that group's total slope.

```
. lincom -2*str-2*1.high#c.str

------------------------------------------------------------------------------
       score | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   3.024396   1.071979     2.82   0.005       .91721    5.131582
------------------------------------------------------------------------------
```

$$\Delta \widehat{Y}_{D=1} = -2(\widehat{\beta}_1 + \widehat{\beta}_3)$$

The SE of a sum needs covariance

Coefficient estimates can move together across samples.

> $\mathrm{Var}(\widehat{\beta}_1 + \widehat{\beta}_3) = V_{11} + V_{33} + 2V_{13}$

> Example: $SE = \sqrt{0.04 + 0.09 + 2(-0.03)} = \sqrt{0.07} \approx 0.265$

Uncertainty of a sum

Try the comparison before opening the explanation.

Practice

In the lesson’s interaction model, the higher-group staffing slope is beta one plus beta three. Can its standard error be obtained by adding the two coefficient standard errors?

Uncertainty of a sum: explanation

Follow the calculation and interpret its units.

> No. Add their variances and twice their covariance, then take the square root. With variances 0.04 and 0.09 and covariance −0.03, the result is the square root of 0.07, about 0.265. Stata’s lincom uses the fitted covariance matrix for the actual model.

Read the higher-group slope as one tested quantity

Use the output for the combination the question asks about.

```
. lincom str+1.high#c.str

------------------------------------------------------------------------------
       score | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |  -1.512198   .5359897    -2.82   0.005    -2.565791    -.458605
------------------------------------------------------------------------------
```

A group gap can depend on X

The group main effect is not the gap at every staffing ratio.

> $\text{Gap}(X) = \beta_2 + \beta_3 X$

> Center at 20: $X^c = X - 20$

A joint hypothesis asks whether several restrictions hold together

One coefficient test cannot replace the combined question.

```
. test 1.high 1.high#c.str

 ( 1)  1.high = 0
 ( 2)  1.high#c.str = 0

       F(  2,   415) =   98.88
            Prob > F =    0.0000
```

> $H_0 : \beta_2 = 0,\ \beta_3 = 0$

> Joint Wald test: evaluate the estimated restrictions using their covariance.

Classical fit-loss F tests require matched models

Restricted and unrestricted fits must answer one nested comparison.

> $$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(n - k_U)}$$

> Example: $F = \frac{(110 - 100)/2}{100/(100 - 5)} = 4.75$

Centering changes coefficient labels, not fitted values

Choose a reference value that helps interpretation.

> $X_i^c = X_i - 20$

Check whether fixed effects absorb a regressor

A time-only variable cannot survive unrestricted time indicators.

> If $Tax_t = 1 + 1\{t = 2\}$, its column duplicates year-intercept variation.

> $Tax_t \times near_i$ can vary across groups within a year.

> A separate coefficient needs variation beyond the included controls.

What year effects absorb

Try the comparison before opening the explanation.

Practice

Tax_t is identical across all districts in each year. Can its coefficient be estimated separately in a regression with an intercept and a full set of year effects? Can $Tax_t \times near_i$ survive?

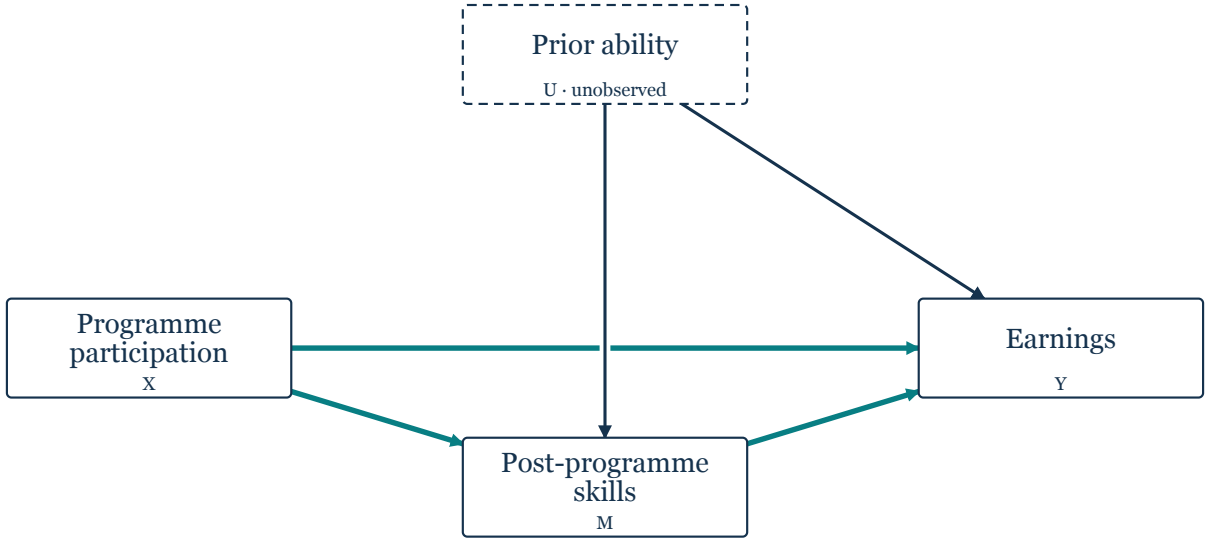
What year effects absorb: explanation

Follow the calculation and interpret its units.

> The common tax column is a linear combination of the year indicators, so its separate coefficient is not identified in that regression. The tax-by-group product can retain variation because it differs across groups within years. Check that it is not reproduced by all the included controls. It describes a differential group slope; it does not recover the absorbed common tax effect.

Separate group heterogeneity from causal identification

A different slope is still a fitted relationship.



2 open directed routes · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Choose the dependence unit for inference

The number of rows need not equal the number of independent sources.

> Cluster: a group within which errors may be related.

> Independent information follows the assignment/error structure, not just the row count.

Build an empirical comparison before reading stars

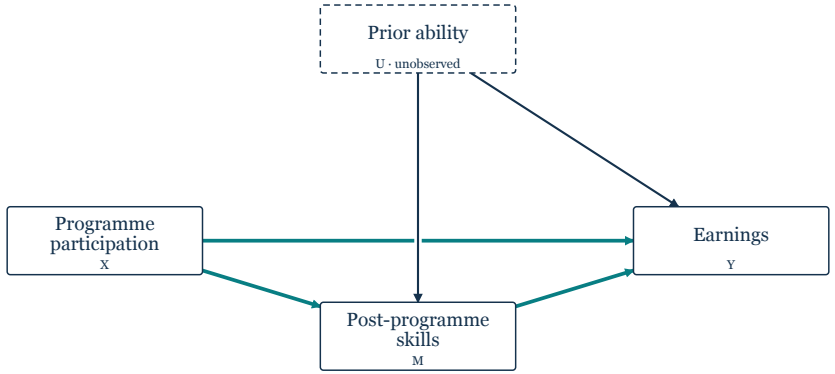
List the coefficient combination and its units.

> For each question, write the required expression: a reference slope, a second-group slope, a slope difference or a group gap at a stated X.

> In a policy memo, report a feasible change within each group and the evidence for any difference between them.

DAG practice: a binary treatment still needs a control argument

Trace the routes before choosing a control.



2 open directed routes · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Practice

Programme participation affects post-programme skills and earnings. Prior ability affects skills and earnings. If the programme is initially independent of ability, what two changes can occur when skills are held fixed? Would adding a group interaction automatically resolve them?

DAG practice: a binary treatment still needs a control argument: explanation

Explain what conditioning changes in this stated graph.

> Conditioning on skills blocks the mediated programme route $X \to M \to Y$. It can also open $X \to M \leftarrow U \to Y$, where skills are a collider and ability influences earnings. A regression can consequently mix a changed target with new selection. A group interaction changes the specified shape or subgroup slope. It does not automatically supply the missing causal control argument. Define the intervention and whether its total or a suitable direct effect is wanted, then assess assignment and mediator–outcome assumptions before interpreting the interaction causally.

Turn an interaction into two policy contrasts

Reproduce the evidence, then explain its economic meaning.

Practice

Centre staffing at 20 and refit the same interaction model. Interpret the resulting group-indicator coefficient and verify that fitted group slopes have not changed. Keep your commands, log, output and interpretation.

Turn an interaction into two policy contrasts: explanation

A reproducible result needs both a calculation and an interpretation.

> Original subgroup staffing slopes: 0.680060 and −1.512198.

> Group gap at ratio 20: −15.429199, robust SE 1.128935.

> Centring changes the reference point, not the predictions.

Can we predict a district left out of fitting?

A prediction question has a different target from a staffing intervention.

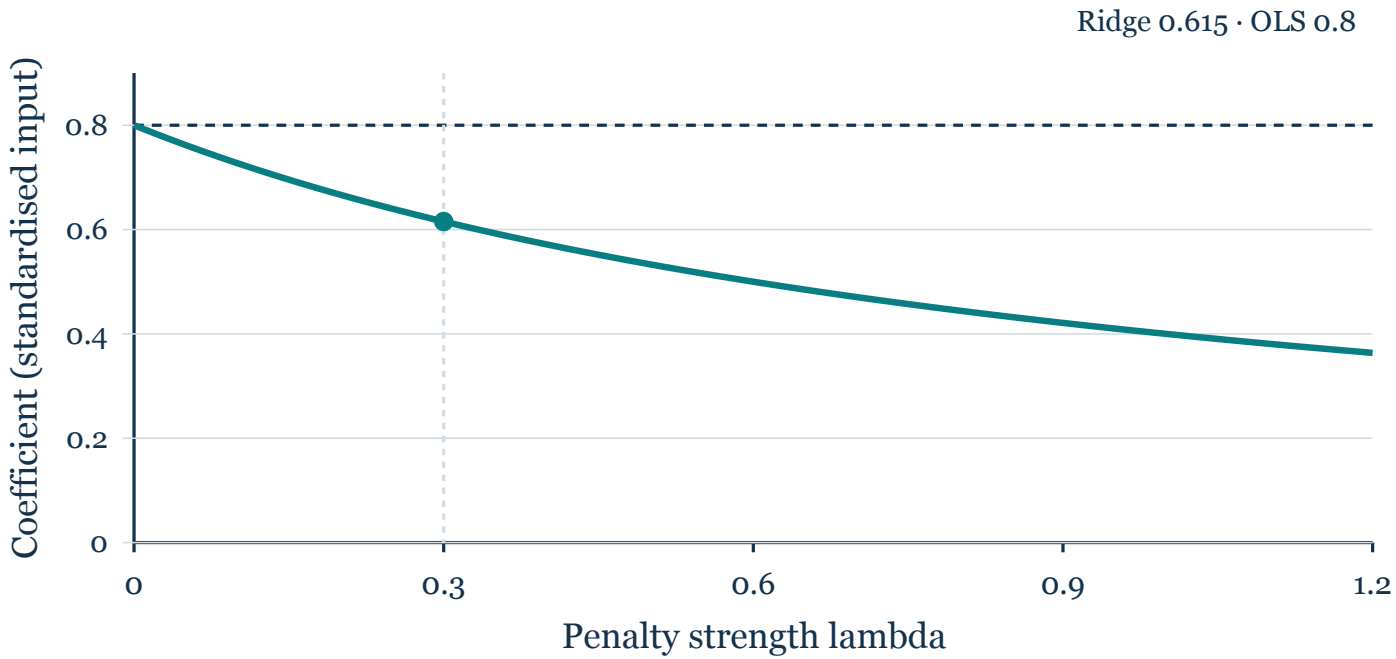
> Training: 280 districts for fitting and tuning.

> Test: 140 districts kept out of fitting and tuning.

> Base: three predictors; expanded: nine predetermined predictor columns.

Ridge shrinks coefficients with a squared penalty

Accept some bias if it reduces enough estimation variation.

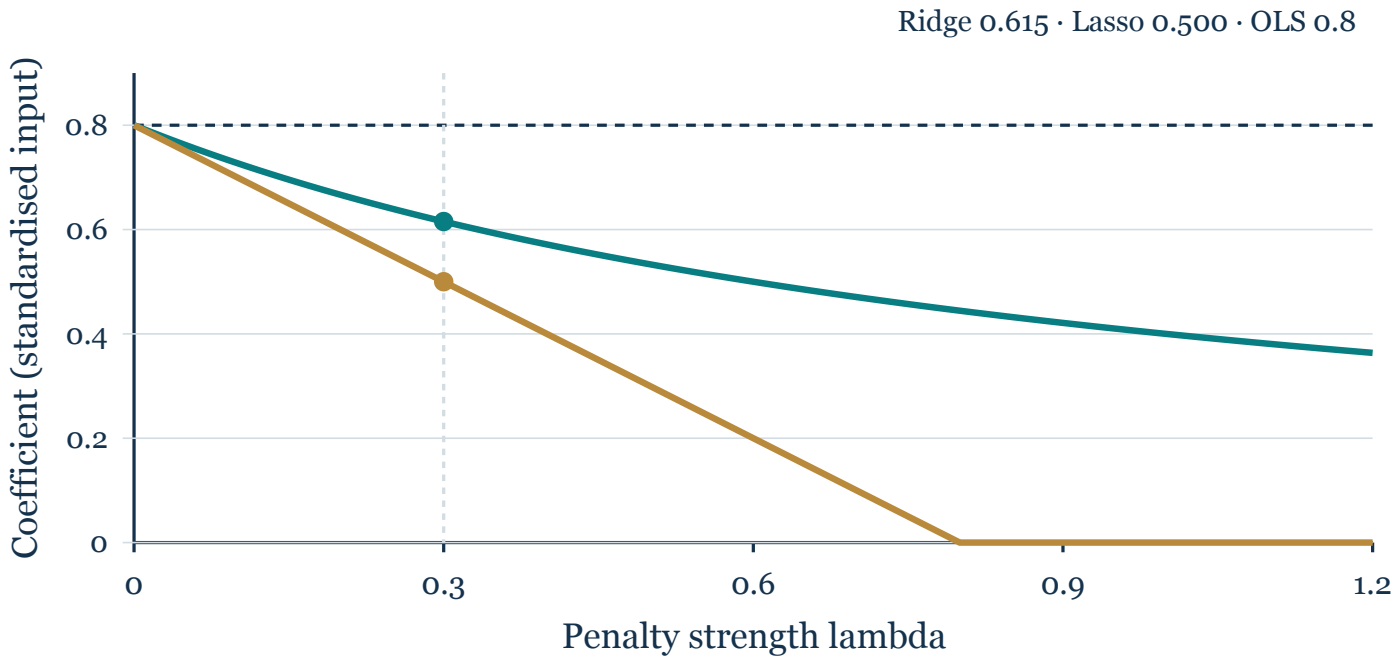


Constructed one-standardised-predictor ridge illustration with OLS slope 0.8. The real nine-column model need not follow this one-column formula.

- Loss: $L(a,b) = \frac{1}{2n}\sum_i [y_i - a - \sum_j b_j z_{ij}]^2$.
- Ridge objective: $L(a,b) + \frac{\lambda}{2}\sum_j b_j^2$.
- Standardise predictor columns using training information; leave the intercept unpenalised.

Lasso can set a predictor coefficient to zero

Absolute and squared penalties behave differently near zero.

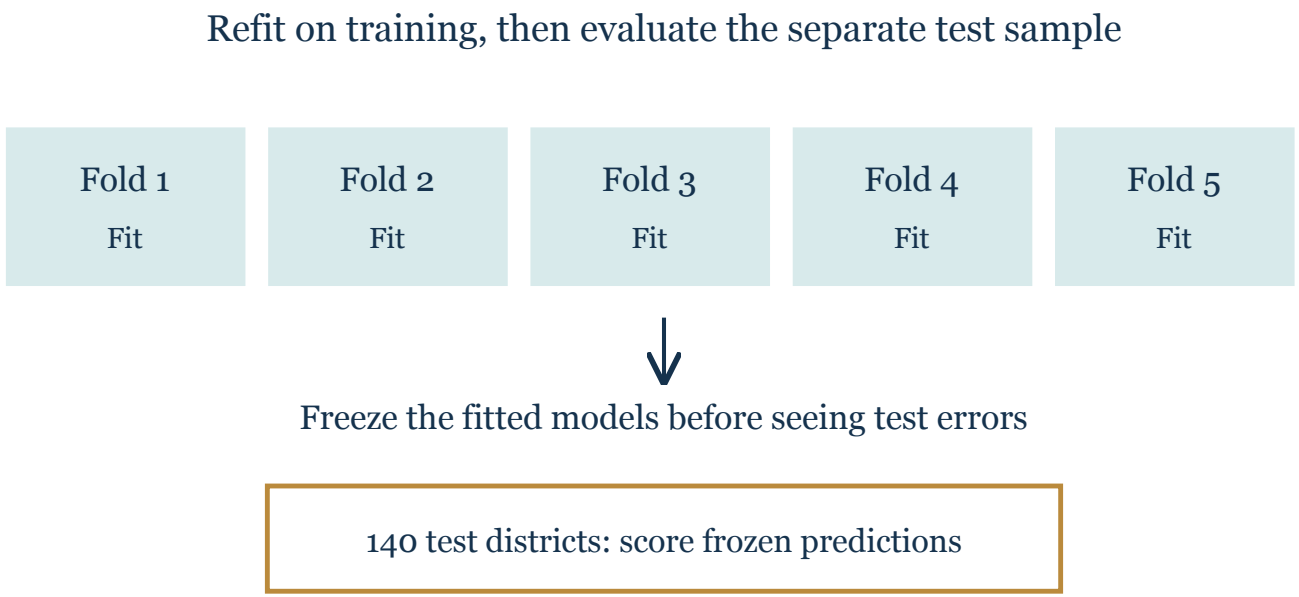


Constructed one-standardised-predictor illustration with OLS slope 0.8. Equal numerical penalties do not imply equal model complexity across methods.

- Lasso objective: $L(a, b) + \lambda \sum_j |b_j|$.
- One standardised predictor: $\hat{b}_R = 0.8/(1 + \lambda)$, $\hat{b}_L = \max(0.8 - \lambda, 0)$.
- These simple paths illustrate the penalties; the actual model has nine correlated columns.

Choose the penalty inside the training sample

Cross-validation reserves the test answers for a later check.

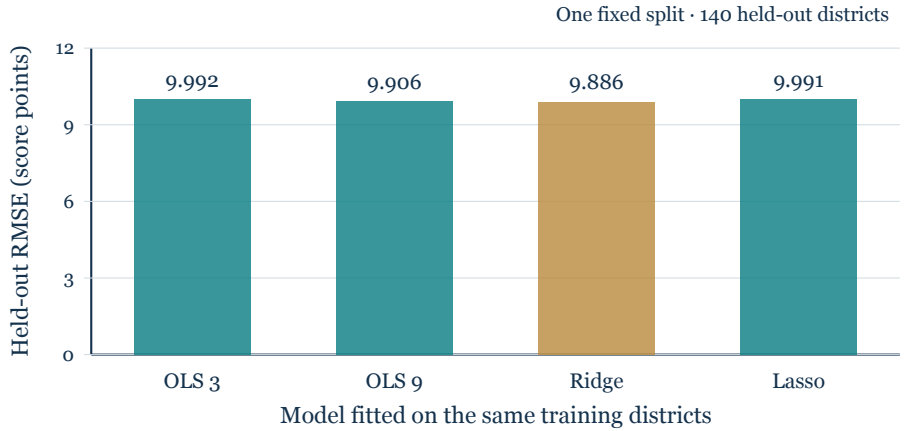


Training-only cross-validation. The separate test sample is used after the fitting and tuning choices are frozen.

- Five training folds: fit on four, validate on one, then rotate.
- Choose the penalty and refit using training observations.
- Score frozen predictions on the separate 140-district test sample.

Compare actual predictions on the same held-out districts

A small error difference is evidence for this run, not a universal ranking.



Actual Stata errors on one fixed split: 280 training and 140 held-out districts. Lower RMSE is better for this prediction task.

- Same training/test split and the same outcome scale for every model.
- Penalties are selected inside training; the test set is evaluated afterwards.
- No test-error ranking identifies a causal staffing effect.

Practice

Does this one test split establish a universally best estimator?

Explanation: No. These small RMSE differences describe this particular evaluation.

Predictive selection does not choose a causal adjustment set

Return to the DAG when the question changes from prediction to intervention.

> Prediction selection depends on the candidate columns, penalty and sample.

> Causal controls follow the target and a defensible identification argument.

> Keep prediction accuracy separate from evidence about an intervention.

Practice

Can prediction lasso justify removing a necessary confounder?

Explanation: No. Causal controls follow the intervention target and identification assumptions.

Carry the same reasoning into ordered observations

Time adds structure to both models and uncertainty.

> We can now represent different group slopes, calculate combined effects and evaluate joint restrictions.

> The next teaching block turns to observations ordered through time.